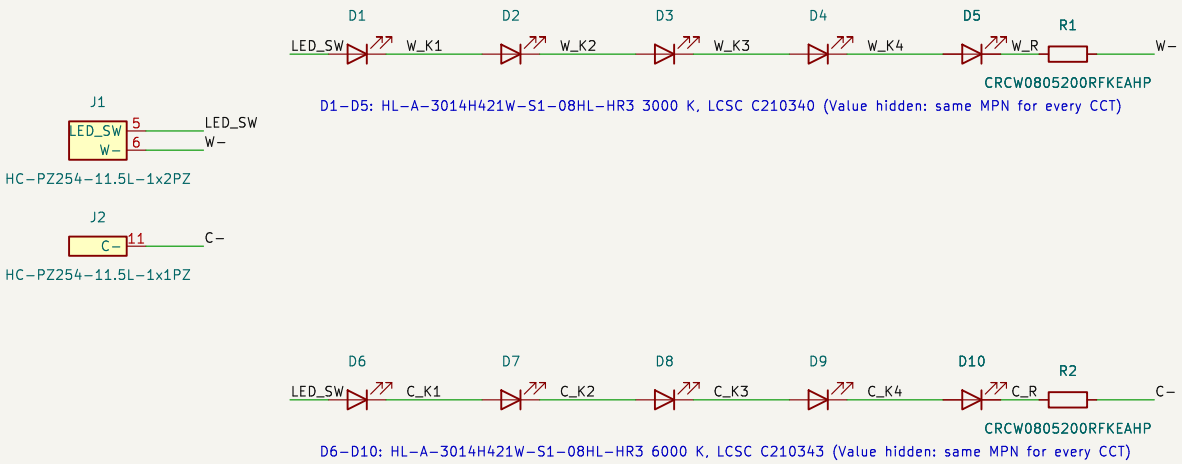


HOW IT CONNECTS (rev B, simple)
J1 (1x2) and J2 (1x1) are standard 2.54 mm straight male headers, hand-soldered by the builder (3 joints).
Pins are numbered by the J6 pin they mate: 5 = LED_SW (anode bus), 6 = W− (warm return), 11 = C− (cool return).
Every other J6 position (incl. J6-12 P+_FUSE and J6-4 GND) has NO pin and no hole on this board.
Header: body 2.5 / post 6.0 / tail 3.0; post tips at Y 108.10 -> J6 engagement 4.72..5.05 mm.

EACH STRING: LED_SW -> 5 LEDs -> Rs 200 R (CRCW0805200RFKEAHP) -> W−/C−. No series diode (rev B):
the housing keys the plug; with 200 R the main board is safe in every misinsertion (research/electrical-j6.md sec 5).
Driver: constant current 13.3 mA (14.0 max), one string at a time; OVP 24.25 V min; design ceiling 22.0 V.
String at 13.3 mA: 5 x Vf 13.0..15.8 V + Rs 2.66 V = LED_SW 16.0..19.4 V (typ 17.3 V, incl. sense + FET) < 22.0 V.
Hot-plug peak through Rs 200 R: TPS923610 47..53 mA, TPS923611 71..77 mA < IFP 80 mA.

LEDs: HONGLITRONIC HL-A-3014H421W-S1-08HL-HR3; the MPN does not encode CCT: warm D1–D5 = LCSC C210340 (3000 K),
cool D6–D10 = LCSC C210343 (6000 K). Pad 1 (large) = cathode. Do not plug the bare board into J6 (no housing = no key).

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Snap-on warm/cool reading light for the Silkscreen reader; plugs into main-board J6
idc LLC

Sheet: /
File: silkscreen-light-attachment.kicad_sch

Title: Silkscreen light attachment

Size: A4	Date: 2026-09-25	**Rev: B**
KiCad E.D.A. 9.0.6		Id: 1/1